Understanding Patronage Patterns in Victoria

Assessment 2: Group Report

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1. Executive Summary

This report presents a data analytics investigation into Victorian public transport patronage on behalf of the Department of Transport and Planning (AusVille). The dataset contains monthly average daily patronage (Pax_daily) across five transport modes: Metrotrain, Tram, Metrobus, Regionaltrain, and Regionalbus, broken down by year, month, day of week, and day type. The goal was to understand how patronage varies by mode and day type, and how these patterns changed across the COVID-19 period.

Key findings:

- Weekday patronage (mean: 267,415 passengers/day) is significantly higher than weekend patronage (mean: 146,640), confirmed by an independent samples t-test ($t = 14.85$, $p < 0.001$).
- Metrotrain accounts for the largest share of total patronage (38%), followed by Tram (33%) and Metrobus (23%).
- Metropolitan modes (Metrotrain, Tram) show declining long-run trends, while regional modes show slight growth or stability.
- The COVID-19 period (2020-2021) caused a sharp drop in patronage across all modes, with only partial recovery observed post-2021.

Recommendation: The Department should focus service recovery efforts on metropolitan modes and plan for growing regional rail demand.

2. Methods

2.1 Requirement Gathering

The dataset provided monthly average daily patronage by day type and transport mode. The key analytical questions were: how does patronage vary by mode, day type, and day of week; and how have patterns changed over time, particularly around the COVID-19 period.

2.2 Data Cleaning and Transformation

All data preparation was performed in Python using the pandas library. The following steps were carried out:

- Type enforcement: Year and Month were cast to integers. Day_of_week, Day_type, and Mode were treated as categorical variables.
- MonthStart column: A proper date column was created from Year and Month (e.g. 2019-03-01) using pd.to_datetime(). This allows time-series ordering and regression modelling.
- Standardisation: All text columns were stripped of whitespace and converted to title-case to remove inconsistencies such as 'TRAIN' vs 'train'.

- Duplicate and null removal: No duplicates or null values were found in the dataset. Defensive dropna() and drop_duplicates() calls were included. Rows with negative Pax_daily values were also removed as physically impossible.
- Feature engineering: Three new columns were added Period (Pre-2020, 2020-2021, Post-2021), Is_Weekday (True for Monday-Friday), and Pax_3m_roll (3-month rolling mean of Pax_daily grouped by Mode and Day_type).
- Export: The cleaned dataset was saved as patronage_clean.csv for use in Power BI.

3. Results

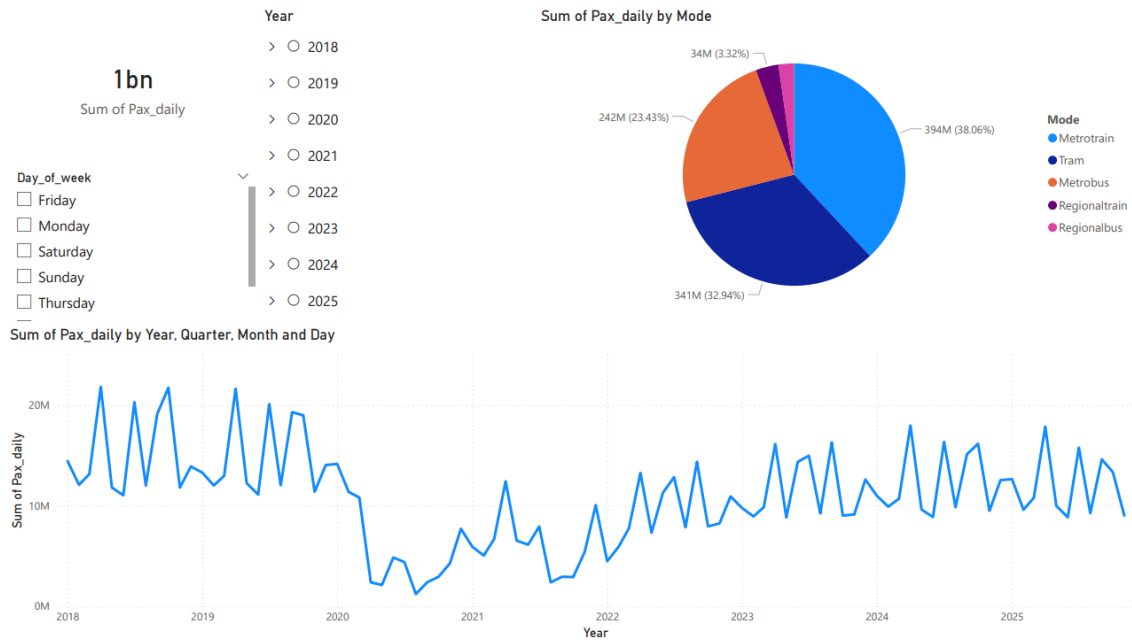
3.1 Descriptive Statistics

The table below summarises average daily patronage and long-run trend by mode:

Mode	Approx. Mean Pax_daily	Regression Slope	Direction
Metrobus	~280,000	+7.28	Slight growth
Metrotrain	~430,000	-41.73	Declining
Regionalbus	~27,000	+1.17	Stable
Regionaltrain	~42,000	+5.21	Growing
Tram	~390,000	-26.59	Declining

3.2 Power BI Dashboard

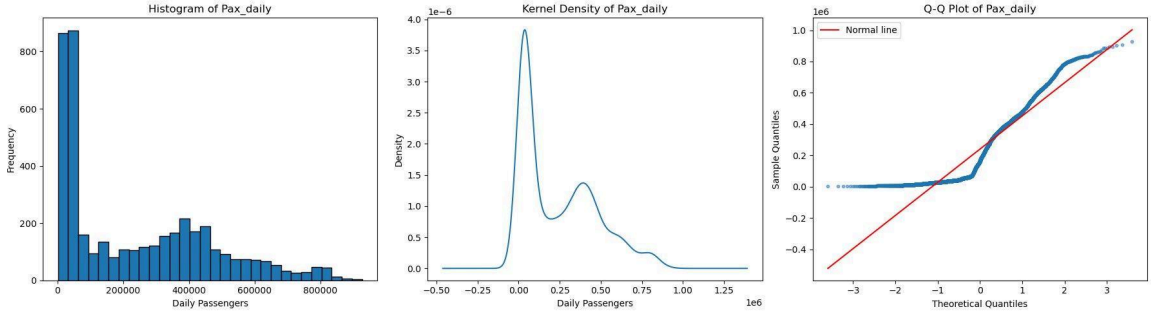
The Power BI dashboard was built using the cleaned dataset and includes an overview page with a pie chart showing patronage share by mode, and a time-series line chart showing total patronage from 2018 to 2025. The dashboard includes year and mode slicers for interactive filtering.



The pie chart confirms that Metrotrain is the dominant mode at 38% of total patronage, followed by Tram at 33% and Metrobus at 23%. Regional modes together account for less than 5% of total system patronage. The time-series line chart clearly shows the COVID-19 shock, a sharp drop beginning in early 2020 and reaching its lowest point in mid-2021, followed by a gradual recovery that has not yet returned to pre-pandemic levels.

3.3 Distribution Analysis

Distribution diagnostics were performed on Pax_daily across the full dataset using a histogram, kernel density estimate (KDE), and Q-Q plot.



The histogram shows a right-skewed, bimodal distribution. The large spike near zero corresponds to low-volume regional services, while the second broader peak around 400,000 represents high-volume metropolitan modes. The KDE plot confirms this bimodal shape.

The Q-Q plot shows significant deviation from the normal reference line, forming an S-shape. This indicates that Pax_daily is not normally distributed across the full dataset. This is expected

because the data is a mixture of modes operating at very different scales. The t-test was still applied on the basis that the large sample size allows the Central Limit Theorem to support approximate normality of sample means.

3.4 Inferential Analysis: Weekday vs Weekend

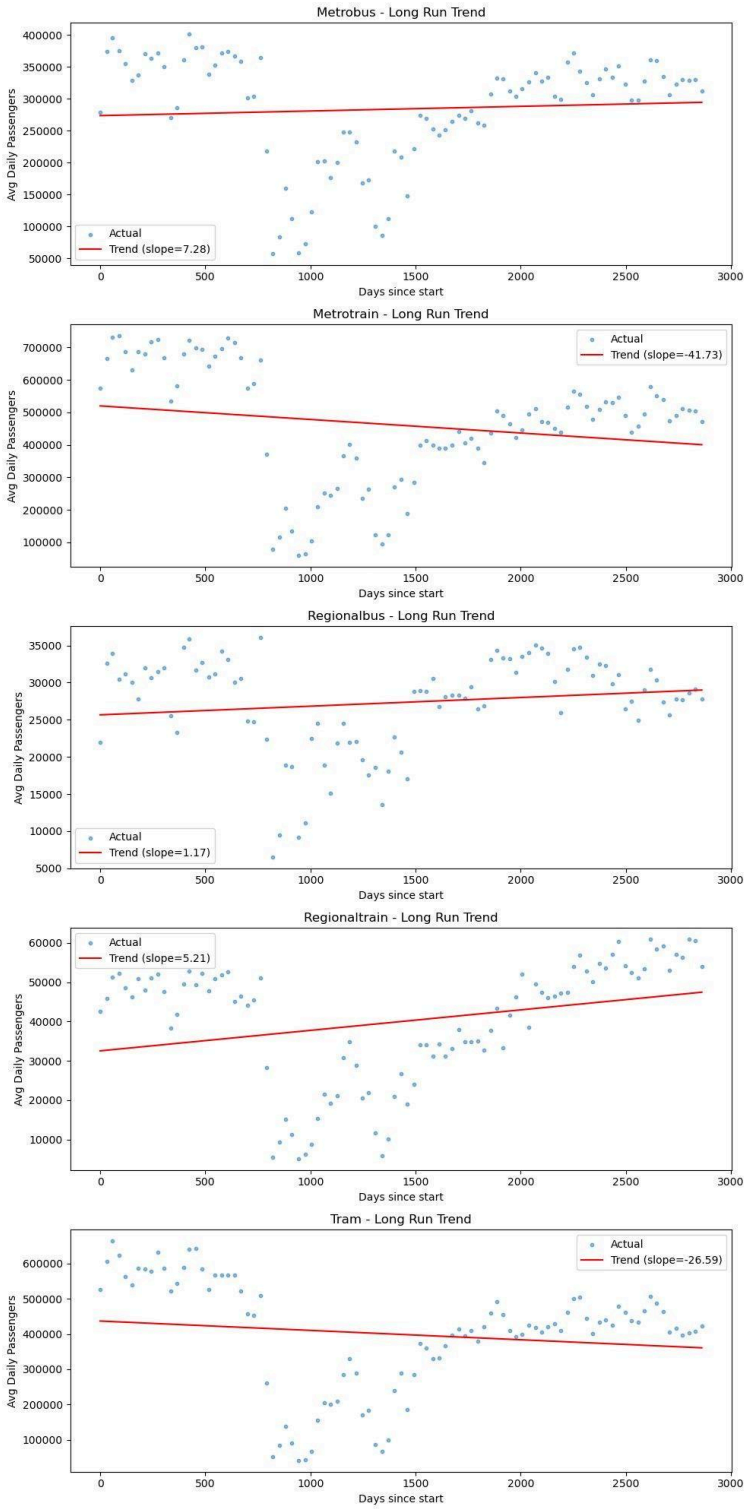
An independent sample t-test was conducted to test whether weekday and weekend patronage differ significantly.

Metric	Weekday	Weekend
Mean Pax_daily	267,415	146,640
T-statistic	14.8527	_
P-value	< 0.0001	_
Conclusion	Significant difference	_

The result is highly significant ($p < 0.001$). Weekday patronage is on average approximately 120,000 passengers per day higher than weekend patronage. This difference has direct implications for timetable planning and staffing levels.

3.5 Long-Run Trend: Linear Regression per Mode

A simple linear regression was fitted per mode with MonthStart (converted to days since the first record) as the predictor and mean Pax_daily as the outcome.



Metrobus (+7.28/day): Broadly stable with slight growth. High monthly variability means the trend is weak.

Metrotrain (-41.73/day): The steepest decline. A severe COVID-era drop is visible and recovery has been incomplete.

Regionalbus (+1.17/day): Essentially flat across the entire study period.

Regionaltrain (+5.21/day): Consistent upward trend, suggesting growing demand for regional rail travel.

Tram (-26.59/day): Second steepest decline, with similar COVID impact to Metrotrain and incomplete post-2021 recovery.

4. Discussion

4.1 Interpretation

The findings show that public transport in Victoria has been significantly affected by the COVID-19 pandemic, and that metropolitan modes have recovered more slowly than regional ones. This may reflect a lasting shift toward working from home, which reduces peak-hour commuter demand for inner-city metro and tram services.

Regional train patronage has been growing steadily throughout the study period, including a relatively faster post-COVID recovery. This could reflect demographic growth in regional areas or a preference shift toward intercity travel. The weekday vs weekend gap remains very large, which confirms the heavily commuter-driven nature of the network.

4.2 Limitations

- The dataset provides monthly averages, not individual trip-level records, which limits granularity.
- Pax_daily is not normally distributed. Non-parametric alternatives such as Mann-Whitney U may be more appropriate for future analyses.
- The linear regression assumes a constant linear trend and cannot capture the non-linear dynamics of COVID recovery.
- No external variables such as fare changes or major infrastructure events are included.

4.3 Implications for Service Planning

- Metrotrain and Tram: Investigate why post-COVID recovery has stalled and consider targeted incentives to win commuters back.
- Regionaltrain: Plan for capacity increases given the consistent upward demand trend.
- Weekday services: Maintain high frequency on weekdays given the statistically confirmed 82% patronage premium over weekends.
- Metrobus and Regionalbus: Stable demand: maintain current service levels in the near term.

5. AI Usage

In accordance with the unit policy, only M365 Copilot was used during this assessment. It was used in the following limited ways:

- Brainstorming report structure: Copilot was prompted to suggest a structure for a data analytics report. The output was reviewed and modified to align with the four required pillars.
- Clarifying statistical concepts: Prompts such as 'what does a Q-Q plot show' were used to verify understanding before applying concepts. No generated text was copied directly.

All submitted text, code, and analysis is the original work of the group. Copilot outputs were used only as a reference aid. Risks such as hallucinations and oversimplifications were mitigated by cross-checking all AI outputs against unit lecture materials before use.